

Name	CSU ID #
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Be sure to read each question carefully. You must choose and answer **exactly two** of the four problems. If you attempt more than two, only the first two will be graded. Write your final answers in the boxes provided. Only answers written in the boxes will be graded. You may use the surrounding space for scratch work. Each problem is worth the same number of points.

1. Let f be continuous. Rewrite

$$\int_0^1 \int_0^{\sqrt{x}} f(x, y) dy dx + \int_1^2 \int_0^{2-x} f(x, y) dy dx$$

as a single double integral in the order $dx dy$. Do not evaluate.

2. A solid lies in the first octant, below $z = 4 - y^2$, with $x + y \leq 2$. Its density is $\rho(x, y, z) = 1 + z$. Set up a triple integral for its mass in the order $dz dy dx$. Do not evaluate.

3. Let h be continuous. Rewrite

$$\int_0^1 \int_x^{2-x} h(x, y) \, dy \, dx$$

in the order $dx \, dy$. Do not evaluate.

4. Let g be continuously differentiable, with $g(0) = 1$ and $g(1) = 4$. Evaluate

$$\int_0^1 \int_0^1 \int_0^{\sqrt{2x}} 2g'(y)z \, dz \, dx \, dy.$$